

Level 3 — Java Exercises

Problem 1 — Leap Year (multiple if/else checks)

Write a LeapYear program that takes a year as input and outputs whether the year is a Leap Year or not. The program should only work for years ≥ 1582 (Gregorian calendar).

Rules:

- Year must be ≥ 1582
- Year is a leap year if it is divisible by 4 and not by 100, unless it is divisible by 400.

```
import java.util.Scanner;

public class LeapYearMultipleIf {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int year = sc.nextInt();
        if (year < 1582) {
            System.out.println("The year " + year + " is not in the Gregorian calendar (year < 1582)");
        } else {
            if (year % 400 == 0) {
                System.out.println("The year " + year + " is a Leap Year");
            } else if (year % 100 == 0) {
                System.out.println("The year " + year + " is not a Leap Year");
            } else if (year % 4 == 0) {
                System.out.println("The year " + year + " is a Leap Year");
            } else {
                System.out.println("The year " + year + " is not a Leap Year");
            }
        }
    }
}
```

Problem 2 — Leap Year (single if using logical operators)

Reimplement the Leap Year check using a single if statement with logical operators `&&` and `||`.

```
import java.util.Scanner;

public class LeapYearSingleIf {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int year = sc.nextInt();
        if (year < 1582) {
            System.out.println("The year " + year + " is not in the Gregorian calendar (year < 1582)");
            return;
        }
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("The year " + year + " is a Leap Year");
        } else {
            System.out.println("The year " + year + " is not a Leap Year");
        }
    }
}
```